

Linear Algebra with Applications

Lecture 9 Subspaces of a Matrix

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1. Introduction: Building on Previous Concepts

In our previous discussions, we established the foundational concepts of vector spaces, linear independence, and matrix multiplication. A key takeaway was interpreting matrix-vector multiplication as a **linear combination of the columns** of the matrix.

The idea of a linear combination is central to linear algebra. Just as we can combine a few basis vectors (like **i** and **j**) to generate an entire vector space (like $\mathbb{R}^2$), the columns of a matrix can be combined to generate their own space. This lecture introduces two fundamental concepts that arise from this idea: the **Rank** of a matrix and its **Column Space**.

2. The Rank of a Matrix

The rank of a matrix is a single number that provides profound insight into its structure and properties.

Definition: Rank of a Matrix

The **rank** of a matrix A is the number of **linearly independent columns** of A . Equivalently, it is also the number of **linearly independent rows** of A . We denote it as $\text{Rank}(A)$.

A common point of confusion is whether to count columns or rows. The remarkable truth is that the result is always the same.

Theorem: Row Rank equals Column Rank

For any matrix A , the number of linearly independent columns is **equal** to the number of linearly independent rows.

$$\text{Rank}(A) = \text{Column Rank}(A) = \text{Row Rank}(A)$$

(Note: The proof for this is a standard result in linear algebra and is an excellent exercise to prove.)

This means that $\text{Rank}(A)$ tells us the "true" dimension of the vector space spanned by its columns (or rows).

Example A Simple 2x2 Matrix

Consider the matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

- **Column View:** The columns are $C_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $C_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. These are the standard basis vectors in $\mathbb{R}^2$ and are clearly **linearly independent**.
- **Conclusion:** Since there are 2 linearly independent columns, **$\text{Rank}(A) = 2$** .

Example A 3x3 Matrix with Dependent Columns

Recall a matrix from our previous lecture:

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 7 & 11 \end{pmatrix}$$

- **Column View:** The columns are $C_1 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$, $C_2 = \begin{pmatrix} 1 \\ 1 \\ 7 \end{pmatrix}$, and $C_3 = \begin{pmatrix} 3 \\ 4 \\ 11 \end{pmatrix}$.
- **Analysis:**
 - C_1 and C_2 are linearly independent (one is not a scalar multiple of the other).
 - Observe that $C_3 = C_1 + C_2$. Since C_3 is a linear combination of C_1 and C_2 , it is linearly **dependent** on them.
- **Conclusion:** The maximum number of linearly independent columns is 2. Therefore, **Rank**(A) = 2.

3. Properties and Implications of Rank

The rank of a matrix is constrained by its dimensions.

Fundamental Property of Rank

The rank of an $m \times n$ matrix A is always less than or equal to the smaller of its dimensions.

$$\text{Rank}(A) \leq \min(m, n)$$

Intuition: You cannot have more linearly independent columns than the total number of columns ($\text{Rank} \leq n$). Similarly, you cannot have more than m linearly independent vectors in an m -dimensional space ($\text{Rank} \leq m$).

3.1. Implication: The "Big Data" Matrix

Imagine a matrix A with dimensions $1000 \times 100,000$ ($m = 1000, n = 100,000$).

- According to the property, $\text{Rank}(A) \leq \min(1000, 100,000)$.
- Therefore, **Rank**(A) ≤ 1000 .

This is a powerful result. Even though there are 100,000 columns (features), at most 1000 of them can be linearly independent. This means that **at least 99,000 columns are redundant**; they can be expressed as linear combinations of the chosen 1000 independent columns. This concept is the cornerstone of **dimensionality reduction** and **data compression**.

4. Subspaces

Before we discuss the spaces generated by a matrix, we need to formalize the idea of a subspace.

Definition: Subspace

A **subspace** is a set of vectors within a larger vector space that is **itself a vector space**.

For a non-empty set S to be a subspace, it must be **closed under linear combinations**. This means for any vectors $\mathbf{v}_1, \mathbf{v}_2 \in S$ and any scalars α, β , the vector $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2$ must also be in S .

Example The x-axis in a 2D plane

- The 2D plane ($\mathbb{R}^2$) is a vector space.
- Consider the set of all vectors on the x-axis, which have the form $\begin{pmatrix} x \\ 0 \end{pmatrix}$.
- Let's check closure: $\alpha \begin{pmatrix} x_1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha x_1 + \beta x_2 \\ 0 \end{pmatrix}$. The result is still on the x-axis.
- **Conclusion:** The x-axis is a subspace of $\mathbb{R}^2$. It is a vector space (a line) that lives inside a larger vector space (a plane).

5. The Column Space of a Matrix

Now, we can formally define the most important subspace associated with a matrix.

Definition: The Column Space (or Range Space)

The **Column Space** of an $m \times n$ matrix A , denoted $\text{Col}(A)$ or $R(A)$, is the set of all possible **linear combinations of the columns of A** .

An equivalent and powerful definition is:

$$\text{Col}(A) = \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y} = A\mathbf{x} \text{ for some vector } \mathbf{x} \in \mathbb{R}^n\}$$

In other words, the column space is the set of all possible outputs of the transformation $T(\mathbf{x}) = A\mathbf{x}$. The Column Space of an $m \times n$ matrix is a **subspace of $\mathbb{R}^m$** .

5.1. Proof that $\text{Col}(A)$ is a Subspace

We will use the closure criteria to prove that the set $\text{Col}(A)$ is a subspace.

1. **Select two arbitrary members:** Let $\mathbf{y}_1$ and $\mathbf{y}_2$ be any two vectors in $\text{Col}(A)$.
2. **Use the definition:** Since $\mathbf{y}_1, \mathbf{y}_2 \in \text{Col}(A)$, there must exist vectors $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$ such that:

$$\mathbf{y}_1 = A\mathbf{x}_1 \quad \text{and} \quad \mathbf{y}_2 = A\mathbf{x}_2$$

3. **Form a linear combination:** We must check if $\alpha\mathbf{y}_1 + \beta\mathbf{y}_2$ is also in $\text{Col}(A)$.

4. Substitute and simplify:

$$\begin{aligned}\alpha \mathbf{y}_1 + \beta \mathbf{y}_2 &= \alpha(A\mathbf{x}_1) + \beta(A\mathbf{x}_2) \\ &= A(\alpha\mathbf{x}_1) + A(\beta\mathbf{x}_2) \quad (\text{Property of matrix multiplication}) \\ &= A(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) \quad (\text{Distributive property})\end{aligned}$$

5. **Analyze the result:** Let $\mathbf{x}_3 = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$. Since $\mathbb{R}^n$ is a vector space, $\mathbf{x}_3$ is also in $\mathbb{R}^n$. Our linear combination is now $A\mathbf{x}_3$.
6. **Conclusion:** Since the result $A\mathbf{x}_3$ is of the form (A times a vector in $\mathbb{R}^n$), it is, by definition, a member of $\text{Col}(A)$. The set is closed under linear combinations and is therefore a **subspace** of $\mathbb{R}^m$.

Practical Applications in Data Science & Machine Learning

- **Dimensionality Reduction & PCA:** A dataset with many features can be represented by a matrix A . If $\text{Rank}(A)$ is low, it means the data has an underlying simple structure. Techniques like Principal Component Analysis (PCA) find an optimal basis for the column space, allowing you to compress massive datasets with minimal information loss.
- **Recommender Systems:** In a user-item rating matrix, a low rank implies that user preferences are driven by a few latent factors (e.g., genres, actors). By finding a basis for the column space, we can uncover these latent factors and make better recommendations.
- **Consistency of Linear Systems:** The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if the vector $\mathbf{b}$ is in the column space of A . In machine learning, this translates to asking: "Can our model A produce this specific output $\mathbf{b}$?" If $\mathbf{b} \notin \text{Col}(A)$, the output is impossible to achieve.

6. Summary & Key Takeaways

- **Rank:** The number of linearly independent columns (or rows). It represents the dimension of the space spanned by the columns/rows. $\text{Rank}(A) \leq \min(m, n)$.
- **Subspace:** A vector space living inside a larger one, closed under linear combinations.
- **Column Space $\text{Col}(A)$:** The set of all possible linear combinations of the columns of A . It is a subspace of $\mathbb{R}^m$ and represents all possible outputs of the transformation $\mathbf{y} = A\mathbf{x}$.

In the next lecture, we will explore another fundamental subspace, the **Null Space**, and see how these two spaces are related.

7. Practice Problems

1. **Calculating Rank:** Find the rank of the following matrix. Justify your answer by showing the linear dependence between the columns.

$$B = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 1 & 5 \\ 2 & 4 & 10 \end{pmatrix}$$

2. **Rank Property:** A matrix C has dimensions 7×4 . What is the maximum possible rank of C ? Why can't it be higher?
3. **Column Space Membership:** Consider the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$ and the vector $\mathbf{b} = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$. Is the vector $\mathbf{b}$ in the column space of A ? In other words, can you find a vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that $A\mathbf{x} = \mathbf{b}$?
4. **Subspace Check:** Is the set of all vectors $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$ where $xy = 0$ a subspace? Explain why or why not by checking the closure properties. (Hint: This set corresponds to the union of the x and y axes).

8. Review and a New Perspective

In our previous lecture, we defined the **Column Space** of a matrix A as the space spanned by its columns and established that its dimension is the **rank** of the matrix.

Review

- **Rank(A):** The number of linearly independent columns of A .
- **Theorem:** For any matrix, Column Rank = Row Rank.
- **Col(A):** The set of all possible outputs $A\mathbf{x}$. It's a subspace of the output space, $\mathbb{R}^m$.

Today, we will deepen our understanding by adopting a new, powerful perspective: viewing a matrix not just as an array of numbers, but as a **geometric transformation**. This view connects the algebraic properties of a matrix (like its rank) to visual, geometric outcomes.

9. Visualizing Matrix Transformations

9.1. Example 1: The Collapsing Transformation (Rank-1 Matrix)

Consider the simple 2×2 matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Let's analyze its effect on a vector $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$.

1. **Calculate the Rank:** The columns are $C_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $C_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Since $C_2 = 1 \cdot C_1$, they are linearly dependent. Thus, **Rank(A) = 1**.
2. **Apply the Transformation:**

$$\mathbf{y} = A\mathbf{x} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x + y \end{pmatrix}$$

3. **Analyze the Geometric Result:** The output vector $\mathbf{y}$ is always of the form $\begin{pmatrix} k \\ k \end{pmatrix}$, which lies on the line $y = x$. This rank-1 matrix takes the **entire 2D plane** and **collapses** it onto a **1D subspace**: the line $y = x$.

9.2. Example 2: Projection from 3D to 2D (Rank-2 Matrix)

Consider the 3×3 matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

1. **Calculate the Rank:** The first two columns are independent; the third is the zero vector. Thus, $\text{Rank}(A) = 2$.
2. **Apply the Transformation:**

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

3. **Analyze the Geometric Result:** This rank-2 matrix takes any point in 3D space and projects it directly down onto the **xy-plane**. The entire 3D input space is mapped to a **2D subspace**, perfectly matching the rank of 2.

10. The Null Space of a Matrix

We've explored the **output** of the transformation $A\mathbf{x}$ (the Column Space). Now, we ask: "Which **input** vectors $\mathbf{x}$ get transformed into the zero vector?"

Definition: The Null Space (or Kernel)

The **Null Space** of an $m \times n$ matrix A , denoted $N(A)$, is the set of all vectors $\mathbf{x}$ in the input space $\mathbb{R}^n$ that are mapped to the zero vector in the output space $\mathbb{R}^m$.

$$N(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$$

The Null Space is a **subspace of the input space** $\mathbb{R}^n$.

Example Finding the Null Space

Let's use our "collapsing" matrix again: $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. We want to find all $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$.

The equation $A\mathbf{x} = \mathbf{0}$ gives us the system $x + y = 0$, or $y = -x$. Any vector on this line can be written as $k \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for some scalar k .

$$N(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

The Null Space of A is the **line** $y = -x$, which is a 1-dimensional subspace of the input space $\mathbb{R}^2$.

10.1. Proof that the Null Space is a Subspace

We formally prove $N(A)$ is a subspace using the closure property. Let $\mathbf{x}_1, \mathbf{x}_2 \in N(A)$. By definition, this means $A\mathbf{x}_1 = \mathbf{0}$ and $A\mathbf{x}_2 = \mathbf{0}$. We check if an arbitrary linear combination

$\alpha \mathbf{x}_1 + \beta \mathbf{x}_2$ is also in $N(A)$.

$$\begin{aligned} A(\alpha \mathbf{x}_1 + \beta \mathbf{x}_2) &= A(\alpha \mathbf{x}_1) + A(\beta \mathbf{x}_2) && \text{(by linearity)} \\ &= \alpha(A\mathbf{x}_1) + \beta(A\mathbf{x}_2) && \text{(by linearity)} \\ &= \alpha(\mathbf{0}) + \beta(\mathbf{0}) \\ &= \mathbf{0} \end{aligned}$$

Since $A(\alpha \mathbf{x}_1 + \beta \mathbf{x}_2) = \mathbf{0}$, the linear combination is in the null space. Therefore, $N(A)$ is a subspace.

11. The Grand Unification: The Rank-Nullity Theorem

We now have two fundamental subspaces for a matrix $A_{m \times n}$:

- **Column Space $\text{Col}(A)$:** A subspace of $\mathbb{R}^m$. Its dimension is $\text{Rank}(A)$.
- **Null Space $N(A)$:** A subspace of $\mathbb{R}^n$. Its dimension is called the **Nullity(A)**.

Theorem: The Rank-Nullity Theorem

For any $m \times n$ matrix A , the dimension of the column space plus the dimension of the null space is equal to the number of columns of the matrix.

$$\text{Rank}(A) + \text{Nullity}(A) = n$$

Intuition: The n dimensions of the input space are "split" by the transformation. Some dimensions are mapped to the column space (the Rank), while the others are "crushed" into the zero vector (the Nullity).

12. Practical Applications in Data Science & ML

Applications of Subspaces

- **Dimensionality Reduction:** Applying a non-square matrix $A_{k \times n}$ (where $k < n$) to a data vector $\mathbf{x} \in \mathbb{R}^n$ is the core idea of dimensionality reduction. The transformation $\mathbf{y} = A\mathbf{x}$ projects high-dimensional data into a lower-dimensional, more manageable space.
- **Feature Engineering:** Understanding which features get mapped to zero (i.e., contribute to the null space) helps us identify and eliminate redundant or irrelevant information. The goal is often to find a transformation A whose null space contains the "noise" or useless features (like "shirt color" for predicting rain).
- **Solving Systems of Equations:** The null space $N(A)$ gives all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$, which is the first step in finding all solutions to the more general problem $A\mathbf{x} = \mathbf{b}$.

13. Summary & Key Takeaways

- **Matrix as a Transformation:** A matrix maps input vectors from $\mathbb{R}^n$ to output vectors in $\mathbb{R}^m$, performing geometric operations like rotation, scaling, and projection.
- **Column Space $\text{Col}(A)$:** The subspace of all possible outputs. Its dimension, the **Rank**, is the dimension of the output space of the transformation.
- **Null Space $N(A)$:** The subspace of all inputs that map to the zero vector. Its dimension is the **Nullity**.
- **Rank-Nullity Theorem:** $\text{Rank}(A) + \text{Nullity}(A) = n$. This connects the dimensions of the input space to the dimensions of the output and null spaces.

14. Practice Problems

1. **Geometric Interpretation:** Consider the matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.
 - (a) Describe what this transformation does to the standard basis vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
 - (b) What is the geometric effect of this transformation on any vector in $\mathbb{R}^2$?
 - (c) What is the rank of this matrix?
2. **Finding Column and Null Space:** Let $A = \begin{pmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \end{pmatrix}$.
 - (a) Determine the rank of A . Describe its column space geometrically.
 - (b) Find a basis for the null space of A by solving $A\mathbf{x} = \mathbf{0}$.
 - (c) What is the nullity of A ?
3. **Verifying Rank-Nullity:** For the matrix A from problem 2, verify that the Rank-Nullity theorem holds.
4. **Conceptual Question:** An $m \times n$ matrix A has $\text{Nullity}(A) = 0$. What does this imply about the solutions to the equation $A\mathbf{x} = \mathbf{0}$? What does it imply about the linear independence of the columns of